



Birzeit University

Computer Science Department

Encryption Theory (Comp 438)

Assignment One

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LSB Algorithm

The **Least Significant Bit (LSB)** approach is a foundational and widely used technique in steganography. It works by replacing the least significant bits of the cover media (such as images or audio files) with bits of secret data to embed information discreetly. Since these LSB alterations minimally impact the media, they are nearly invisible to human perception, making the hidden message difficult to detect.

To illustrate, in an 8-bit image with pixel values ranging from 0 to 255, the LSB of each pixel can be modified to store a bit of the secret message. For example, if a pixel has a value of 110 (binary: 01101110), and we want to embed a bit of 1, we can replace the last bit to make it 01101111. This tiny change is often imperceptible, particularly in large cover files like high-resolution images or audio files, ensuring that the original content appears unchanged to the human eye or ear.

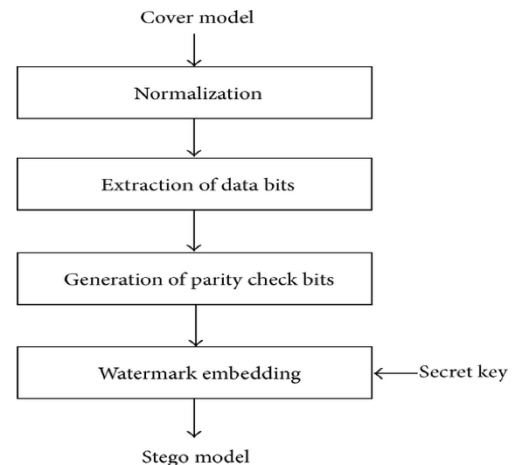


Figure 1: Flow Diagram of Embedding Images.

For example:

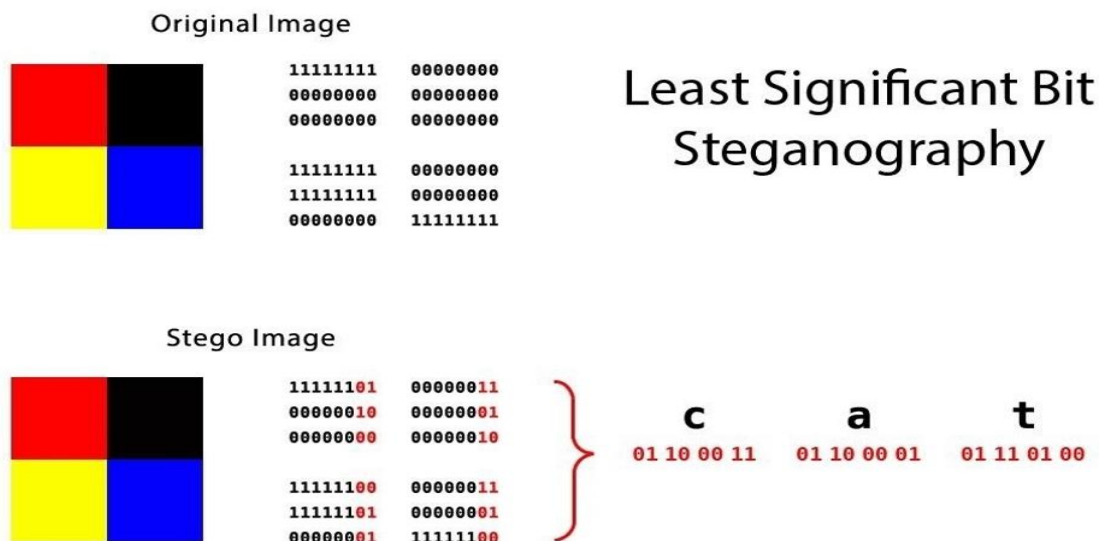


Figure 2: Least Significant Bit Steganography.

Chosen Distribution Technique: Block-Based Distribution

In this implementation, a **block-based sequential channel distribution** technique is chosen to embed the hidden message within an image. Instead of sequentially embedding data across individual pixels, this approach divides the image into distinct, non-overlapping blocks, with each block exclusively assigned to a group of pixels' values within a single-color channel (Red, Green, or Blue). The distribution proceeds in a sequential manner across blocks: the first blocks embed data in the green channel of image pixels, the second blocks do so in the red channel, and the third ones in the blue channel, continuing in this pattern throughout the image.

Within each block, the Least Significant Bit (LSB) of each selected color channel is modified to store parts of the hidden message. This controlled, sequential block-based channel assignment ensures that no single color channel is modified across the entire image, which minimizes visible patterns and enhances security. By using distinct blocks and rotating channels in a set sequence, the technique prevents easily detectable modifications and makes it difficult for unauthorized parties to discern the hidden data without knowledge of the block structure and channel rotation pattern.

Impact of Distribution Technique on Detectability

In this implementation, data is hidden by manipulating the three color channels (Red, Green, and Blue) within each pixel, but each block focuses on a group of pixels' values within a specific channel. Specifically, we chose to have four pixels per block, with each block embedding only one bit of the message at each iteration: the first blocks embed data only in the green channels of those pixels, the following ones use only the red channels, and the last blocks use only the blue channels. If all blocks of all channels have been used such that each of them embeds a single bit, they will be reused in next iterations, where in each iteration, every block will embed a new single bit until the whole message gets hidden in the image. This controlled use of color channels enhances security and reduces detectability by distributing message bits selectively across different channels and blocks.

Mechanism of Action

1. **Selective Channel Embedding:** Each pixel in an RGB image contains three channels (Red, Green, and Blue). Rather than modifying all channels uniformly, this approach divides the embedding work across blocks, with each block

containing data only in one specific color channel. For example, within a four-pixel block, one block might exclusively use the green channels across all four pixels, another block might embed in the red channels, and one block in the blue. This structured separation within blocks prevents any single color channel from being consistently altered, minimizing detectable patterns.

2. **Non-Corresponding Message Encoding Across Channels:** To further enhance security, data within the green, red, and blue channels of the same pixel does not correspond to the same character or part of a word. For instance, if a pixel's green channel represents part of one word, the red and blue channels in that same pixel may represent different words or sections of the message. This technique helps obscure the message, making it more challenging for any unauthorized party to extract meaningful data even if they detect channel modifications. The inconsistency in message encoding within each pixel disrupts attempts to decipher content based on channel similarities, adding a layer of misdirection.
3. **Sequential Channel Rotation:** The LSBs of the selected color channels are modified in a rotating sequence, so each block has a unique embedding pattern that further obfuscates the hidden message. This technique takes advantage of the fact that each channel has different sensitivities to color perception, reducing the risk of noticeable color shifts. The controlled channel rotation across blocks makes the message nearly imperceptible, as each color's contribution to the visual output remains subtle.

Impact on Detectability

By combining block distribution with selective channel embedding and non-corresponding message encoding, this method significantly reduces the risk of statistical detection. Even if forensic analysis is applied, the patterns within a single color channel remain minimal due to the staggered channel use and the non-uniform representation of the message. This setup complicates any attempt to extract the hidden message, as it requires knowledge of both the block structure and the specific channel rotations and divisions. The approach effectively distracts and misleads potential attackers, ensuring a robust defense against unauthorized decryption.

Robustness Evaluation of the Block-Based Distribution

This method is designed to be very secure due to the way it distributes hidden data in the image.

1. **Rotating Channels Across Blocks:** The data is hidden in a rotating sequence across the Red, Green, and Blue channels within different blocks of the image. Spreading data across channels makes it harder for anyone trying to find patterns or detect hidden information. Even if someone notices small changes in the pixels, it would be difficult to piece together the hidden message without knowing the exact sequence used.
2. **Different Parts of the Message in Each Channel:** Within each pixel, the Red, Green, and Blue channels hold different parts of the message, meaning no single pixel contains a full word or character. This mixing acts as a distraction, so even if someone identifies modified pixels, they wouldn't be able to reconstruct the message by examining just one channel or pixel. This adds complexity, requiring detailed knowledge of the encoding structure to decode the message.
3. **Scattering Data with Block Division:** By dividing the image into separate blocks, each holding data in only one color channel, the message is widely scattered. Without knowing the block size, the rotation sequence, and which channels hold data, it would be nearly impossible to extract the message. This structure protects the data from being decoded by brute-force methods, as someone would have to guess the exact embedding setup to retrieve the message.
4. **Reducing Detection of Patterns:** This block-based approach reduces the chance of detectable patterns, which are often found in simpler LSB techniques. By avoiding clustering of altered bits in one color, it prevents analysis tools from spotting hidden data based on unusual color patterns.

In summary, without knowing the exact details of how the data was embedded—like the block layout and the channel rotation—extracting the hidden message would be very difficult. This technique combines both spatial and color-based distribution to keep the data secure and protected from unauthorized access or decryption.

A Comparison between the Block-Based Distribution and Sequential Embedding Techniques

Message Security

Block-Based Distribution: This technique enhances security by scattering the message across blocks with specific color channel rotations (green, red, blue) in non-overlapping segments. This prevents a single channel or section from consistently carrying message bits, making the embedded data difficult to detect via color analysis. The staggered channel use, combined with varying block locations, minimizes patterns, raising the difficulty for attackers attempting to identify the hidden message.

Sequential Embedding: Sequential methods, which embed data continuously across pixels and channels, are more vulnerable to pattern recognition. Because data is embedded in a predictable, linear fashion, sequential embedding may expose the message location through statistical analysis or pixel intensity shifts, making it easier for potential intruders to identify and possibly extract the hidden data.

Impact on Image Quality

Block-Based Distribution: By targeting a specific channel per block and rotating through channels, the block-based method achieves subtle data embedding with minimal visual disruption, distributing modifications evenly. Consequently, the image quality is preserved, reducing the risk of noticeable artifacts that might suggest tampering.

Sequential Embedding: As sequential embedding modifies a continuous series of bits, it can cause detectable artifacts, especially when high amounts of data are embedded. This can result in quality degradation that could reveal the presence of hidden information if the embedding intensity is too high.

Robustness

Block-Based Distribution: This approach's robustness stems from the random-like dispersion and controlled channel rotation, which complicates unauthorized extraction. Without knowing the exact block size, channel rotation sequence, or data location, decoding becomes highly challenging. The added complexity effectively thwarts brute-force extraction efforts, as an attacker would need to determine multiple parameters to reconstruct the message accurately.

Sequential Embedding: While simpler to implement, the sequential method's predictability makes it more vulnerable to extraction attempts. A skilled attacker could more readily analyze sequentially embedded data patterns, making extraction feasible with fewer guesses than required for block-based methods.

The block-based distribution technique thus provides a secure and robust alternative to sequential embedding, with improved image quality and resistance against detection and extraction attempts.

Now here the original image and the steganography image after hide this message inside it: "This is a test for block distribution technique used in LSB steganography".

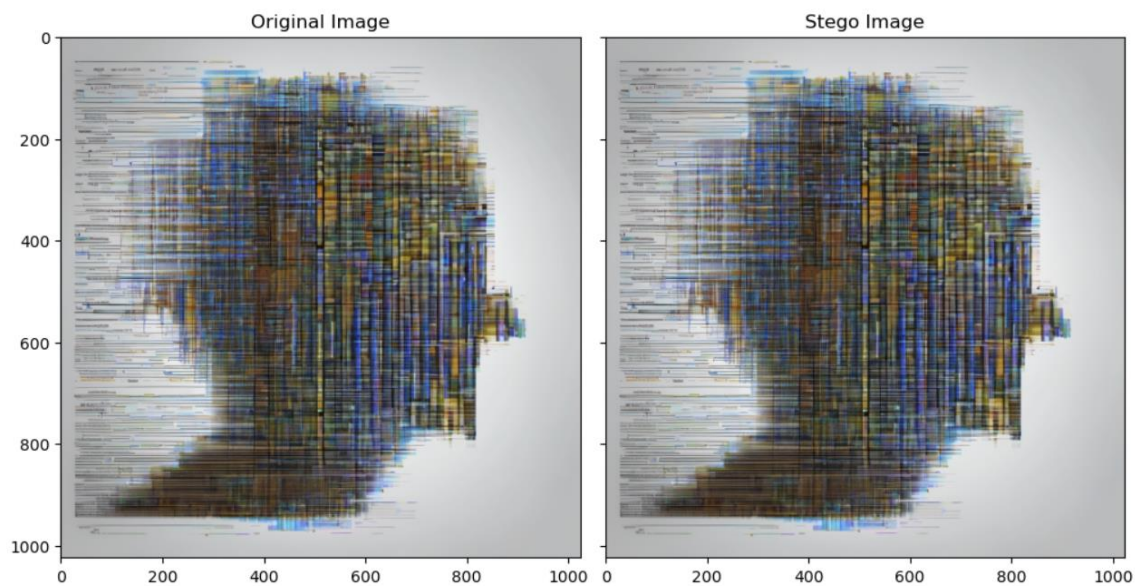


Figure 3: Original Cover Image Vs. Stego-Image.

The image used in this technique employs **lossless compression**, meaning it retains its original quality regardless of how many times it's saved, opened, or resized. This high fidelity ensures that even with hidden data embedded, the image quality remains consistent. As a result, any potential attacker would be unable to detect the presence of a hidden message based solely on visual or quality degradation, as the image will still appear unaltered. This use of lossless compression further enhances the security of the embedded message by maintaining the original appearance and quality of the image.